

HIMITSU PROTOCOL

Swap from a private note

Private note → Uniswap V2 → new private note

PROTOCOL PRESENTATION · 2026



THE PROBLEM

Wallet swaps expose the link to a trade

- / Wallet address is public
- / Swap amount and timing are public
- / Private notes need a constrained path to AMM liquidity

THE SOLUTION

Swap from one private note to another



/ Proof authorizes a limited spend

/ Swap output goes into a new private note

EIP-8141 · TYPE 0X06

One transaction, four frames

01	02	03	04
Expiry	Pool verify	Paymaster	Pool send
Reject expired transaction	Check proof and spend	Approve gas payment	Swap and deposit output

/ Pool is the transaction sender

/ Prefunded paymaster pays network gas

WHY THIS DESIGN

Different transaction paths, different trade-offs

APPROACH	PATH	TRADE-OFF
Direct wallet	User sends and pays	Wallet is the public sender
Relayer	Service signs and submits	Adds service dependency
ERC-4337	UserOperation · EntryPoint · bundler	Separate account-abstraction route
EIP-8141	Frames in type 0x06	Pool sender · separate paymaster

/ Frames combine validation, payment, and execution

/ ERC-4337 also supports sponsored gas

RELATED PRIVACY PROTOCOLS

Himitsu focuses on one private swap route

PROTOCOL	FOCUS	DIFFERENCE
Tornado Cash Classic	Private deposits and withdrawals	No integrated AMM output-note flow
RAILGUN	Shielded balances and private calls	Broader private DeFi
Himitsu	EIP-8141 note spend through Uniswap V2	Output committed to a second note

/ Narrower transaction path

/ Not a replacement for broader privacy systems

PRIVACY BOUNDARIES

The proof hides the note, not the swap

Private

- / Spent note commitment
- / Note secret and witness
- / Proof generated in browser

Public

- / Amounts and timing
- / AMM activity
- / Recipient and RPC metadata

CURRENT BUILD · NEXT VALIDATION

Working prototype, with network and security work ahead

Built

- / Browser notes and Groth16 proofs
- / EIP-8141 frame transaction
- / Uniswap V2 swap to output note

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Next

- / Public compatible network
- / Independent contract and circuit review
- / Measure privacy and user experience